IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MATHEMATICS: PRINCIPLES AND ENGINEERING APPLICATIONS B

Monday, 27 April 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions. Question 1 & 2 are Mathematics, worth 30 & 45 marks, question 3 is Signals and Communications, worth 25 marks.

NO CALCULATORS ALLOWED.

Any special instructions for invigilators and information for candidates are on page 1.

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THE QUESTIONS

1. a) A periodic function is defined as

$$h(t) = \begin{cases} 1-t & \text{for} \quad 0 \le t < 1 \\ 2t-1 & \text{for} \quad 1 \le t < 2 \end{cases}$$

and $h(t+2) = h(t)$ for all t. A Fourier series is obtained for this function. [*You do not need to do this!*] What are the values of the Fourier series at $t = 0$ and $t = 3$? [3]

b) A function $f(t)$ is defined as

$$f(t) = \begin{cases} 0 & \text{for} \quad 0 \le t < \frac{1}{2} \\ t - \frac{1}{2} & \text{for} \quad \frac{1}{2} \le t < 1 \, . \end{cases}$$

Let $g_1(t)$ and $g_2(t)$ be the even and odd extensions, respectively, of $f(t)$, both with period $T = 2$. Sketch g_1 and g_2, for $-3 \le t \le 3$. [3]

c) Obtain the Fourier Transform $F(\omega)$, of the function

$$f(t) = \begin{cases} \cos(t) \, , & \text{for} \quad -\pi \le t \le \pi \, , \\ 0 \, , & \text{otherwise} \end{cases}$$

in terms of $\sin(\pi\omega)$. [4]

d) Obtain the cartesian equation of the plane through the points $A(1,2,-3)$, $B(0,1,2)$ and $C(3,4,-5)$. [3]

e) Find the distance from the point $P(1,-2,2)$ to the plane with cartesian equation $-2x+2y+z=5$. Find the point Q on the plane, which is nearest to P. [4]

f) Use Gaussian elimination to find the inverse of the matrix

$$\begin{pmatrix} 3 & -1 & 2 \\ -4 & 1 & -2 \\ 1 & 1 & 1 \end{pmatrix}. \qquad [4]$$

g) Given

$$A = \begin{pmatrix} -1 & 3 & 2 \\ 1 & 1 & 1 \\ 1 & -4 & -2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 & -2 \\ 2 & -2 & -1 \\ -3 & 1 & 2 \end{pmatrix},$$

obtain $\det(A)$, $\det(B)$ and $\det(AB)$. [3]

h) Find the stationary points of the function

$$f(x,y) = y(x+2)^2 + y^2 - 4y\,. \qquad [3]$$

i) Classify the stationary points of the function

$$f(x,y) = y(x+2)^2 + y^2 - 4y\,,$$

which you found in the previous question. [3]

2. a) Obtain the first five non-zero terms of the Fourier Series of $g_1(t)$, the period-2 even extension of $f(t)$ from question 1(b):

$$f(t) = \begin{cases} 0 & \text{for} \quad 0 \le x < \frac{1}{2} \\ t - \frac{1}{2} & \text{for} \quad \frac{1}{2} \le x < 1 \end{cases}.$$

[8]

b) In question 1(h) and (i), you found and classified the stationary points of $f(x,y) = y(x+2)^2 + y^2 - 4y$. Now use these to sketch the contours of the function. [5]

c) Given the matrix

$$A = \begin{pmatrix} 2 & -2 & 3 \\ -1 & 1 & -2 \\ 3 & 2 & 4 \end{pmatrix};$$

i) Use Gaussian elimination to show that A cannot be written in the form $A = LU$ where L is lower triangular and U upper triangular. Hence, identify the permutation matrix P needed so that $LU-$decomposition can be done on PA;

ii) Use Gaussian elimination to transform PA into the upper triangular U and write down the elimination matrices needed for this;

iii) Write down the inverses of the elimination matrices;

iv) Hence, or otherwise, find the corresponding lower triangular L, so that $PA = LU$.

[8]

d) Using Laplace transforms, or otherwise, solve

$$\frac{d^2x}{dt^2} + 4x = \begin{cases} 4 & \text{for} \quad 0 \le t \le 1 \\ 0 & \text{for} \quad t > 1 \end{cases}$$

and $x(0) = 0, \quad x'(0) = 1$, writing out the solution explicitly for the intervals $0 < t < 1$ and $t > 1$. [7]

e) Given the linear system of equations $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ where

$$A = \begin{pmatrix} 1 & 1 & 1 & 4 \\ 1 & 0 & 2 & 1 \\ 1 & -3 & 5 & -8 \end{pmatrix},$$

i) For $\underline{\mathbf{b}} = \begin{pmatrix} 1 \\ -1 \\ \alpha \end{pmatrix}$, obtain the echelon form of the augmented coefficient matrix.

ii) Identify free and pivot columns and free and pivot variables; state the rank of A.

iii) Use the free variables to find the special solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$. What does each special solution say about the columns of A?

iv) Give the nullspace and column space of A.

v) Find a condition on α to guarantee solvability of $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$. Hence, obtain the general solution.

[10]

f) The following solution of a first-order ordinary differential equation (ODE) is incorrect.

The first order ODE

$$\left(x^2 + \frac{1}{y} + e^y\right)\frac{dy}{dx} + 2xy + \frac{1}{x} = 0 \tag{2.1}$$

can be seen to be exact by performing the standard check. We write the solution as $f(x,y) = $ constant, and compute

$$\frac{\partial f}{\partial x} = 2xy + \frac{1}{x} \quad \Rightarrow \quad f = x^2 y + \ln(x) + g(y). \tag{2.2}$$

From this and the original equation we have

$$\frac{\partial f}{\partial y} = x^2 + g' = x^2 + \frac{1}{y} + e^y \tag{2.3}$$

which gives that

$$g = \ln(y) + e^y. \tag{2.4}$$

Hence the solution is

$$x^2 y - e^y + \ln\left(\frac{x}{y}\right) = constant. \tag{2.5}$$

Which was the *first* error made in this calculation? Eliminate the error and obtain the correct solution. [7]